

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – IX

Roll No.:T045	Name:Omkar Thukarul
Class:TYIT	Batch:1
Date of Assignment: 18-08-2026	Date/Time of Submission:18-08-2026

Part B - Program

- a. Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.

CODE:

Program.cs:-

```
using Microsoft.EntityFrameworkCore;
using P9A.Data;
public partial class Program
{
    private static void Main(string[] args)
    {
        var builder = WebApplication.CreateBuilder(args);
        builder.Services.AddControllers();
        builder.Services.AddDbContext<AppDbContext>(options =>
            options.UseSqlServer(
                builder.Configuration.GetConnectionString("DefaultConnection")));
        builder.Services.AddEndpointsApiExplorer();
        builder.Services.AddSwaggerGen();
        var app = builder.Build();
        app.UseSwagger();
        app.UseSwaggerUI();
        app.UseHttpsRedirection();
        app.UseAuthorization();
        app.MapControllers();
        app.Run();
    }
}
```

Models.cs:-

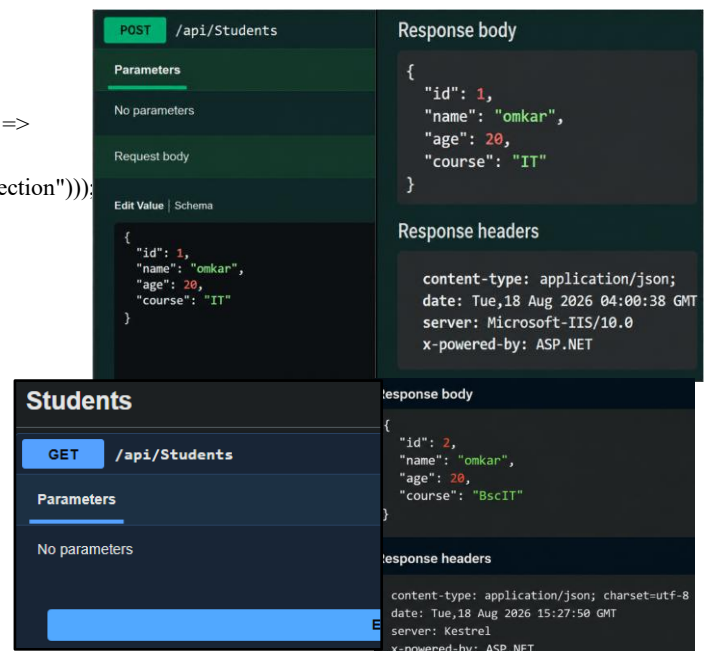
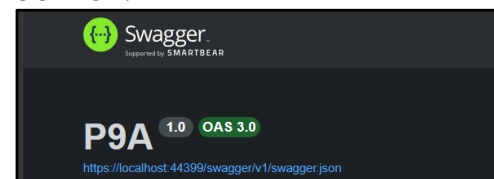
namespace P9A.Models

```
{
    public class Student
    {
        public int Id { get; set; }
        public required string Name { get; set; }
        public int Age { get; set; }
        public required string Course { get; set; }
    }
}
```

AppDbContext.cs

```
using Microsoft.EntityFrameworkCore;
using P9A.Models;
namespace P9A.Data
{
    public class AppDbContext : DbContext
```

OUTPUT:



Max Rows: 1000			
Id	Name	Age	Course
1	omkar	20	IT
2	omkar	20	BscIT

```

{
    public AppDbContext(DbContextOptions
<AppDbContext> options)
: base(options)
    {
    }

    public DbSet<Student> Students { get; set; }
}

```

appsettings.json:-

```

"ConnectionStrings": {
    "DefaultConnection": "Server=(localdb)\\mssqllocaldb;
Database=P9A;Trusted_Connection=true;"
},

```

StudentController.cs:-

```

using Microsoft.AspNetCore.Mvc;
using P9A.Data;
using P9A.Models;
namespace P9A.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        private readonly AppDbContext _context;

        public StudentsController(AppDbContext context)
        {
            _context = context;
        }

        // GET ALL
        [HttpGet]
        public IActionResult GetStudents()
        {
            return Ok(_context.Students.ToList());
        }

        // GET BY ID
        [HttpGet("{id}")]
        public IActionResult GetStudent(int id)
        {
            var student = _context.Students.Find(id);

            if (student == null)
                return NotFound();

            return Ok(student);
        }

        // INSERT
        [HttpPost]
        public IActionResult AddStudent(Student student)
        {
            _context.Students.Add(student);
            _context.SaveChanges();
            return Ok(student);
        }

        // UPDATE
        [HttpPut("{id}")]
        public IActionResult UpdateStudent(int id, Student student)
        {

```

GET /api/Students/{id}

Parameters

Name	Description
id * required	integer(\$int32) (path)

Execute

Response body

```
{
  "id": 2,
  "name": "omkar",
  "age": 20,
  "course": "BscIT"
}
```

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 18 Aug 2026 15:27:50 GMT
server: Kestrel
x-powered-by: ASP.NET
```

Id	Name	Age	Course
2	omkar	20	IT
88	NULL	NULL	NULL

PUT /api/Students/{id}

Parameters

Name	Description
id * required	integer(\$int32) (path)

Execute

Request body

```
{
  "id": 0,
  "name": "Omkar",
  "age": 21,
  "course": "MCA"
}
```

Responses

Curl

```
curl -X 'PUT' \
'https://localhost:7047/api/Students/2' \
-H 'accept: */*' \
-H 'Content-Type: application/json' \
-d '{
  "id": 0,
  "name": "Omkar",
  "age": 21,
  "course": "MCA"
}'
```

Request URL

```
https://localhost:7047/api/Students/2
```

Id	Name	Age	Course
2	Omkar	21	MCA

Responses

Curl

```
curl -X 'DELETE' \
'https://localhost:44399/api/Students/1' \
-H 'accept: */*' \
-H 'Content-Type: application/json'
```

Request URL

```
https://localhost:44399/api/Students/1
```

Server response

Code	Details
200	Response headers

```
date: Tue, 18 Aug 2026 04:04:54 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
```

Responses

Code	Description
200	OK

DELETE /api/Students/{id}

Parameters

Name	Description
id * required	integer(\$int32) (path)

Execute

dbo.Students [Data]

Id	Name	Age	Course
88	NULL	NULL	NULL

```

var data = _context.Students.Find(id);
if (data == null)
    return NotFound();
data.Name = student.Name;
data.Age = student.Age;
data.Course = student.Course;
_context.SaveChanges();
return Ok(data);
}
// DELETE

[HttpDelete("{id}")]
public IActionResult DeleteStudent(int id)
{
    var student = _context.Students.Find(id);
    if (student == null)
        return NotFound();
    _context.Students.Remove(student);
    _context.SaveChanges();
    return Ok();
}
}
}

```

b. Create an API endpoint to calculate Factorial of a Number and test it using Postman.

CODE:

MathController.cs:-

using Microsoft.AspNetCore.Mvc;

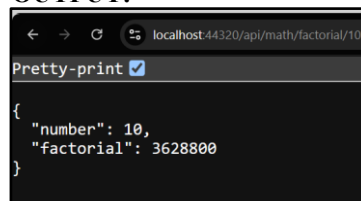
namespace Fact.Controllers

```

{
    [Route("api/[controller]")]
    [ApiController]
    public class MathController :
    ControllerBase
    {
        [HttpGet("factorial/{number}")]
        public IActionResult GetFactorial(int number)
        {
            if (number < 0)
                return BadRequest
                ("Factorial not possible for negative numbers.");
            long factorial = 1;
            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }
            return Ok( new
            {
                Number = number,
                Factorial = factorial
            });
        }
    }
}

```

OUTPUT:



```

{
  "number": 10,
  "factorial": 3628800
}

```

